

Numbers were the input, numbers were the output.

Babbage is such an important figure, a few links must be mentioned.

For historical documents:

www.fourmilab.ch/babbage/contents.html

For lots of links:

www.zyvex.com/nanotech/babbage.html

For a biography:

[www-history.mcs.st andrews.
ac.uk/Biographies/Babbage.html](http://www-history.mcs.st-andrews.ac.uk/Biographies/Babbage.html)

For lots of interesting stuff:

[http://ei.cs.vt.edu/~history/
Babbage.html](http://ei.cs.vt.edu/~history/Babbage.html)



Behind the mechanics, the mathematics, the machine: Charles Babbage, who started it all

1822

Charles Babbage designed his first mechanical computer, a prototype for his Difference Engine. One must remember that Babbage invented two machines—the Analytical Engine and the Difference Engine (cf. Müller, 1786). Both were too complex to actually be built. The Analytical Engine, which Babbage outlined in 1833, used punched cards for input—very interestingly similar to the early “real” computers.

1832

Babbage, in collaboration with Joseph Clement, an engineer and draughtsman, produced a prototype segment of the Difference Engine, which operated on six-digit numbers, and which could tabulate quadratic polynomials. (A polynomial is something like $a^2 + b$; a quadratic polynomial is something like $a^4 + b^3 - c^2 + d$.) The Engine was planned to be the size of a room. The output digits, it was planned, would be punched onto a soft metal plate; a printing press could use it, and voila—readable output!